

# SRINIVAS BIYAGUDA

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## Summary

Senior Software Engineer with over 3 years of experience at **Tata Consultancy Services**, specializing in full-stack development with advanced proficiency in **Python, PySpark, big data tools**, and **SQL databases**. Expertise in designing scalable microservices, processing large-scale datasets, and optimizing data pipelines for performance and security. Adept at mentoring teams, driving innovation, and delivering high-impact solutions in Agile environments, with a focus on scalability, efficiency, and collaboration.

## Education

### Clemson University

*Master of Science in Computer Science, GPA: 3.80/4.00*

### Sreenidhi Institute of Science and Technology

*Bachelor of Technology in Computer Science and Engineering, GPA: 9.51/10.00*

**Clemson, SC**

*Aug 2023 – May 2025*

**Hyderabad, India**

*July 2016 – Aug 2020*

## Technical Skills

**Programming Languages:** Python (Advanced), PySpark, JavaScript, TypeScript, Java, C++

**Big Data Tools:** Apache Spark, Hadoop, Kafka, Hive

**Frontend Technologies:** React, Angular, Vue.js, HTML, CSS, SASS, Tailwind CSS

**Backend Technologies:** Node.js, Express.js, Django, Spring Boot

**Databases:** SQL, MySQL, PostgreSQL, MongoDB, DynamoDB

**Cloud and DevOps:** AWS (Lambda, S3, EC2, RDS), Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions)

**Software Practices:** Agile, Test-Driven Development (TDD), SOLID Principles, Design Patterns

**Version Control:** Git, GitHub

**Security:** OWASP, OAuth, JWT, SAML

## Professional Experience

### Tata Consultancy Services

*Senior Software Engineer*

**Aug 2020 – Jun 2023**

*Hyderabad, India*

- Developed scalable **microservices** using **Python, PySpark**, and **Spring Boot** for Deutsche Bank, processing over **100 million structured records** with **99.9% uptime**.
- Optimized **big data pipelines** using **Apache Spark** and **Hadoop**, reducing data processing time by **35%** for CitiBank.
- Designed and implemented **RESTful APIs** with **Node.js** and **Django**, integrated with **PostgreSQL** and **MySQL**, improving transaction throughput by **20%**.
- Built reusable front-end components using **React** and **Angular**, streamlining development and reducing UI development time by **25%**.
- Deployed applications on **AWS** (EC2, Lambda, S3) using **Docker** and **Kubernetes**, enhancing scalability and reducing deployment time by **15%**.
- Mentored junior developers in **Agile** practices and **TDD**, improving team productivity by **10%**.
- Ensured security compliance with **OWASP** standards and **JWT** authentication, mitigating vulnerabilities in production systems.

### Triant Software Solutions

*Software Engineering Intern*

**Jan 2020 – Jul 2020**

*Hyderabad, India*

- Developed and optimized web applications using **React** and **Node.js**, reducing page load times by **30%**.
- Created **RESTful APIs** and integrated third-party services, enhancing user experience and system interoperability.
- Contributed to **CI/CD pipelines** using **Jenkins** and **GitHub Actions**, streamlining deployment processes.
- Participated in system design discussions, evaluating trade-offs for scalability and security using **Python** and **SQL**.

## Projects

### Secure E-commerce Platform

*Python, PySpark, React, AWS*

**2022**

- Developed a full-stack e-commerce platform using **Python** and **PySpark** for backend data processing, handling **10,000 transactions daily**.
- Implemented a responsive front-end with **React** and **Tailwind CSS**, improving user engagement by **20%**.
- Integrated secure authentication (**JWT**) and payment gateways, ensuring compliance with **OWASP** standards.
- Deployed on **AWS** (RDS, S3, Lambda), optimizing scalability with **Docker** and **Kubernetes**.

### Scalable Microservices Architecture

*Python, Django, PostgreSQL, Kubernetes*

**2021**

- Designed **microservices** using **Python** and **Django**, integrated with **PostgreSQL**, supporting **1 million requests/day**.
- Utilized **PySpark** for real-time data analytics, reducing query execution time by **40%**.
- Orchestrated deployment with **Kubernetes**, ensuring zero-downtime updates and high availability.

- Implemented **CI/CD** pipelines with **GitHub Actions**, accelerating release cycles by **15%**.

## Open-Source Contributions

2020 – 2023

### *Python, JavaScript, Big Data Tools*

- Contributed to **Apache Spark** and **PySpark** libraries, enhancing big data processing capabilities.
- Developed features for open-source **JavaScript** frameworks, improving community-driven solutions.
- Optimized **SQL** queries in open-source database tools, increasing performance by **10%**.

## Certifications

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- NPTEL Certified in Data Science, Python, C Programming
- Google Code Jam Certified in Problem Solving, Data Structures & Algorithms
- Udemy Certified in Full-Stack Web Development with ReactJS
- Security Training (OWASP, Secure Coding Practices)